# JEE–MAIN EXAMINATION – JANUARY 2026

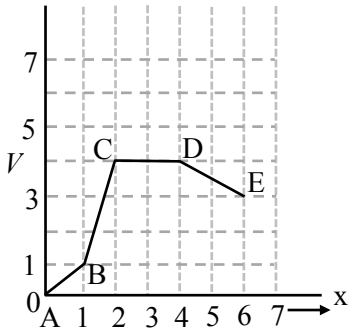
**(HELD ON WEDNESDAY 21st JANUARY 2026)** **TIME : 9:00 AM TO 12:00 NOON**

## PHYSICS

## TEST PAPER WITH ANSWER

### SECTION-A

**26.** Potential energy ($V$) versus distance ($x$) is given by the graph, Rank various regions as per the magnitudes of the force (F) acting on a particle from high to low.




(1) $F_{BC} > F_{CD} > F_{DE} > F_{AB}$

(2) $F_{CD} > F_{AB} > F_{BC} > F_{DE}$

(3) $F_{CD} > F_{DE} > F_{AB} > F_{BC}$

(4) $F_{BC} > F_{AB} > F_{DE} > F_{CD}$

**Ans. (4)**

**27.** A gas based geyser heats water flowing at the rate of 5.0 litres per minute from 27°C to 87°C.

The rate of consumption of the gas is ____ g/s.

(Take heat of combustion of gas = $5.0 \times 10^4$ J/g) specific heat capacity of water = 4200 J/kg. °C

(1) 2.1 (2) 4.2

(3) 0.42 (4) 0.21

**Ans. (3)**

**28.** A conducting circular loop of area 1.0 $m^2$ is placed perpendicular to a magnetic field which varies as B = sin(100 t) Tesla. If the resistance of the loop is 100 Ω , then the average thermal energy dissipated in the loop in one period is _____J.

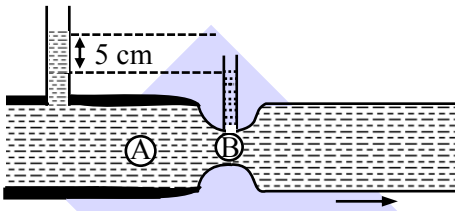
(1) $\frac{\pi}{2}$ (2) $2\pi$

(3) $\pi$ (4) $\pi^2$

**Ans. (3)**

**29.** Water flows through a horizontal tube as shown in the figure. The difference in height between the water columns in vertical tubes is 5 cm and the area of cross-sections at $A$ and $B$ are 6 $cm^2$ and 3 $cm^2$ respectively.

The rate of flow will be_____$cm^3/s$. (take g = 10 $m/s^2$)




(1) $\frac{200}{\sqrt{3}}$ (2) $200\sqrt{6}$

(3) $200\sqrt{3}$ (4) $100\sqrt{3}$

**Ans. (3)**

**30.** In an experiment the values of two spring constants were measured as $k_1 = (10 \pm 0.2)$ N/m and $k_2 = (20 \pm 0.3)$ N/m. If these springs are connected in parallel, then the percentage error in equivalent spring constant is :

(1) 2.67% (2) 2.33%

(3) 1.33% (4) 1.67%

**Ans. (4)**

**31.** A 4 kg mass moves under the influence of a force $\vec{F} = (4t^3\hat{i} - 3t\hat{j})$ N where t is the time in second.

If mass starts from origin at t = 0, the velocity and position after t = 2s will be :

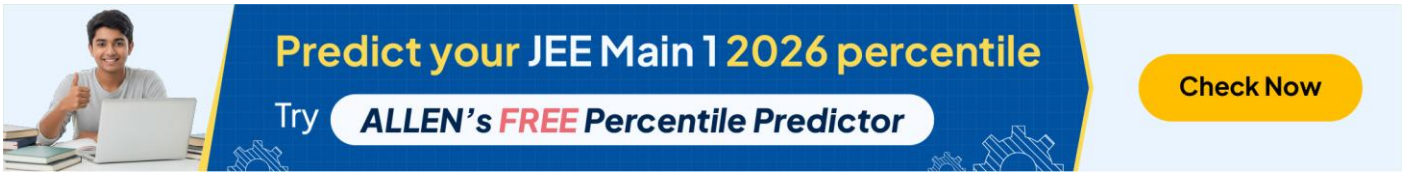
(1) $\vec{v} = 3\hat{i} + \frac{3}{2}\hat{j}$ $\vec{r} = \frac{6}{5}\hat{i} + \hat{j}$

(2) $\vec{v} = 4\hat{i} - \frac{3}{2}\hat{j}$ $\vec{r} = \frac{8}{5}\hat{i} - \hat{j}$

(3) $\vec{v} = 4\hat{i} + \frac{5}{2}\hat{j}$ $\vec{r} = \frac{8}{5}\hat{i} + 2\hat{j}$

(4) $\vec{v} = 4\hat{i} - \frac{3}{2}\hat{j}$ $\vec{r} = \frac{6}{5}\hat{i} - \hat{j}$

**Ans. (2)**



**32.** Consider a modified Bernoulli equation.

$$\left(P+\frac{A}{Bt^2}\right)+\rho g(h+Bt)+\frac{1}{2}\rho V^2=\text{constant}$$

If t has the dimension of time then the dimensions of A and B are _____, _____respectively.

(1) $[ML^0T^{-1}]$ and $[M^0LT]$

(2) $[ML^0T^{-1}]$ and $[M^0LT^{-1}]$

(3) $[ML^0T^{-2}]$ and $[M^0LT^{-2}]$

(4) $[ML^0T^{-2}]$ and $[M^0LT^{-1}]$

**Ans. (2)**

**33.** A current carrying is placed vertically and a particle of mass m with charge Q is released from rest. The particle moves along the axis of solenoid. If g is acceleration due to gravity then the acceleration (a) of the charged particle will satisfy :

(1) a = g (2) a > g

(3) a = 0 (4) 0 < a < g

**Ans. (1)**

**34.** A parallel plate capacitor has capacitance C, when there is vacuum within the parallel plates.

A sheet having thickness $\left(\frac{1}{3}\right)^{rd}$ of the separation between the plates and relative permittivity K is introduced between the plates. The new capacitance of the system is :

(1) $\frac{3KC}{2K+1}$ (2) $\frac{CK}{2+K}$

(3) $\frac{3CK^2}{(2K+1)^2}$ (4) $\frac{4KC}{3K-1}$

**Ans. (1)**

**35.** The electric field a plane electromagnetic wave is given by :

$E_y = 69\sin[0.6\times10^3x - 1.8\times10^{11}t]$ V/m.

The expression for magnetic field associated with this electromagnetic wave is_____T.

(1) $B_z = 2.3\times10^{-7}\sin[0.6\times10^3x - 1.8\times10^{11}t]$

(2) $B_z = 2.3\times10^{-7}\sin[0.6\times10^3x + 1.8\times10^{11}t]$

(3) $B_y = 69\sin[0.6\times10^3x + 1.8\times10^{11}t]$

(4) $B_y = 2.3\times10^{-7}\sin[0.6\times10^3x - 1.8\times10^{11}t]$

**Ans. (1)**

**36.** In a double slit experiment the distance between the slits is 0.1 cm and the screen is placed at 50 cm from the slits plane. When one slit is covered with a transparent sheet having thickness t and refractive index n(= 1.5), the central fringe shifts by 0.2 cm. The value of t is ____ cm.

(1) $8\times10^{-4}$ (2) $6.0\times10^{-3}$

(3) $5.6\times10^{-4}$ (4) $5.0\times10^{-3}$

**Ans. (1)**

**37.** A light wave described by $E = 60\sin(3\times10^{15})t + \sin(12\times10^{15})t]$ (in SI units) falls on a metal surface of work function 2.8 eV. The maximum kinetic energy of ejected photoelectron is (approximately) _____ eV. ($h = 6.6\times10^{-34}$ J-s. and $e = 1.6\times10^{-19}$C)

(1) 5.1 (2) 3.8

(3) 6.0 (4) 7.8

**Ans. (1)**

**38.** If an alpha particle with energy 7.7 MeV is bombarded on a thin gold foil, the closest distance from nucleus it can reach is _____ m.

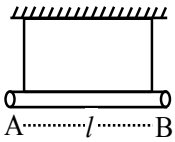
(Atomic number of gold = 79 and $\frac{1}{4\pi\epsilon_0} = 9\times10^9$ in SI units)

(1) $2.95\times10^{-14}$ (2) $2.95\times10^{-16}$

(3) $3.85\times10^{-16}$ (4) $3.85\times10^{-14}$

**Ans. (1)**

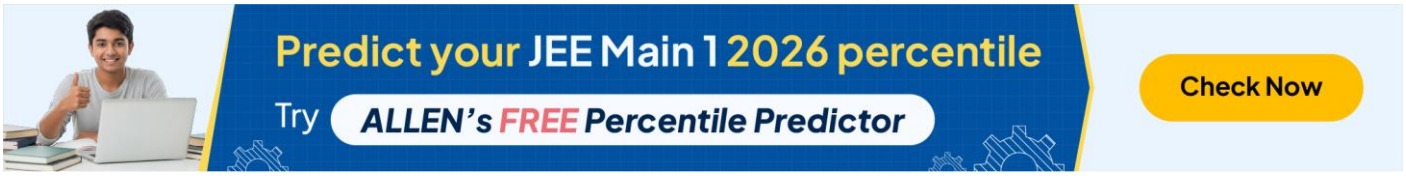
**39.** A uniform rod of mass m and length *l* suspended by means of two identical inextensible light strings as shown in figure. Tension in one string immediately after the other string is cut, is _____. (g acceleration due to gravity)




(1) mg/2 (2) mg/4

(3) mg/3 (4) mg

**Ans. (2)**



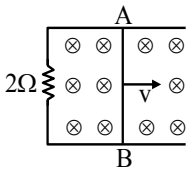
**40.** An aluminium and steel rods having same lengths and cross-sections are joined to make total length of 120 cm at 30°C. The coefficient of linear expansion of aluminium and steel are $24 \times 10^{-6}$/°C and $1.2 \times 10^{-5}$/°C, respectively. The length of this composite rod when its temperature is raised to 100°C, is _____ cm.

(1) 120.20 (2) 120.15

(3) 120.03 (4) 120.06

**Ans. (2)**

**41.** A 1 m long metal rod AB completes the circuit as shown in figure. The area of circuit is perpendicular to the magnetic field of 0.10 T. If the resistance of the total circuit is 2Ω then the force needed to move the rod towards right with constant speed (v) of 1.5 m/s is _____ N.

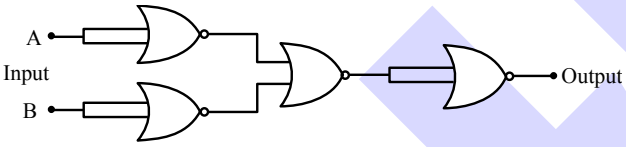



(1) $7.5 \times 10^{-2}$ (2) $5.7 \times 10^{-3}$

(3) $5.7 \times 10^{-2}$ (4) $7.5 \times 10^{-3}$

**Ans. (4)**

**42.** The given circuit works as :




(1) AND gate (2) NOR gate

(3) NAND gate (4) OR gate

**Ans. (3)**

**43.** Two strings (A, B) having linear densities $\mu_A = 2 \times 10^{-4}$ kg/m and $\mu_B = 4 \times 10^{-4}$ kg/m and lengths $L_A$ = 2.5 m and $L_B$ = 1.5 m respectively are joined. Free ends of A and B are tied to two rigid supports C and D, respectively creating a tension of 500 N in the wire. Two identical pulses, sent from C and D ends, take time $t_1$ and $t_2$, respectively, to reach the joint. The ratio $t_1/t_2$ is :

(1) 1.08 (2) 1.90

(3) 1.67 (4) 1.18

**Ans. (4)**

**44.** Initially a satellite of 100 kg is in a circular orbit of radius $1.5R_E$. This satellite can be moved to a circular orbit of radius $3R_E$ by supplying $\alpha \times 10^6$J of energy. The value of α is _____.

(Take Radius of Earth $R_E = 6 \times 10^6$ m and g = 10 m/s$^2$)

(1) 150 (2) 500

(3) 100 (4) 1000

**Ans. (4)**

**45.** A point charge of $10^{-8}$ C is placed at origin. The work done in moving a point charge 2 μC from point A(4, 4, 2) m to B(2, 2, 1) m is _____J.

$(\frac{1}{4\pi \in_0} = 9 \times 10^9$ in SI units)

(1) $45 \times 10^{-6}$

(2) 0

(3) $30 \times 10^{-6}$

(4) $15 \times 10^{-6}$

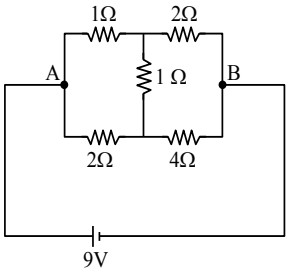
**Ans. (3)**

## SECTION - B

**46.** A collimated beam of light of diameter 2 mm is propagating along x-axis. The beam is required to be expanded in a collimated beam of diameter 14 mm using a system of two convex lenses. If first lens has focal length 40 mm, then the focal length of second lens is _____ mm.

**Ans. (280)**

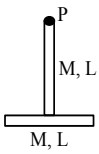
**47.** The heat generated in 1 minute between points A and B in the given circuit, when a battery of 9V with internal resistance of 1 Ω is connected across these points is _____ J.




**Ans. (1080)**



**48.** Two identical thin rods of mass M kg and length L m are connected as shown in figure. Moment of inertia of the combined rod system about an axis passing through point P and perpendicular to the plane of the rods is $\frac{x}{2}ML^2$ kg m$^2$. The value of x is _____.




**Ans. (17)**

**49.** 10 mole of oxygen is heated at constant volume from 30°C to 40°C. The change in the internal energy of the gas is _____ cal. (The molecular specific heat of oxygen at constant pressure, $C_p$ = 7 cal./mol °C and R = 2 cal./mol °C.)

**Ans. (500)**

**50.** In a microscope the objective is having focal length $f_0$ = 2 cm and eye-piece is having focal length $f_e$ = 4 cm. The tube length is 32 cm. The magnification produced by this microscope for normal adjustment is _____.

**Ans. (100)**



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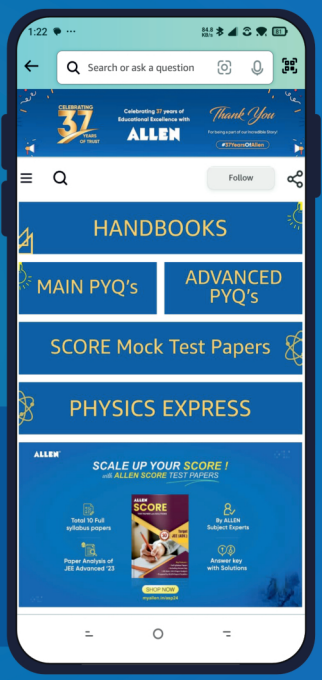
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